

2022

LONG QUESTIONS

- ① A Survey of 200 families known to be regular television viewers was undertaken. They were asked which of 3 television channels they watch most during an average week. A summary of their replies is given in the following table together with the region in which they lived:

Channel	Region			
	North	East	South	West
PTV 1	29	46	42	23
PTV 2	6	11	26	7
STN	15	3	12	10

Test the hypothesis There is no association between channel watched most and the region

Solution:-

Hypothesis:-

H_0 : There is no association between the channel watched most and the region.

H_1 : There is association between the channel watched most and the region.

Level of Significance:-

$$\alpha = 0.05$$

Test-Statistics:-

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

Calculation:-

Chennel	Region				Total
	North	East	South	west	
PTV 1	29	42	23		110
PTV 2	6	11	26	7	50
STN	15	3	12	10	40
Total	50	30	80	40	200

observed frequency o_i	expected frequency e_i	$(o - e_i)$	$(o - e_i)^2$	$\frac{(o - e_i)^2}{e_i}$
29	27.5	1.5	2.25	0.0818
6	12.5	-6.5	42.25	3.38
15	10	5	25	2.5

16	16.5	-0.5	0.25	0.01515
11	7.5	3.5	12.25	1.633
3	6	-3	9	1.3
42	44	-2	4	0.09090
26	20	6	36	1.8
12	16	-4	16	1
23	22	1	1	0.04545
7	10	3	9	0.9
10	8	2	4	0.5
200	200	---	---	$\chi^2 = 13.2463$

Find expected Frequency e_i

$$\frac{50 \times 10}{200} = 2.5 \quad \frac{80 \times 110}{200} = 44$$

$$\frac{50 \times 50}{200} = 12.5 \quad \frac{80 \times 50}{200} = 20$$

$$\frac{50 \times 40}{200} = 10$$

$$\frac{80 \times 40}{200} = 16$$

$$\frac{30 \times 110}{200} = 16.5$$

$$\frac{40 \times 110}{200} = 22$$

$$\frac{30 \times 50}{200} = 7.5$$

$$\frac{40 \times 50}{200} = 10$$

$$\frac{30 \times 40}{200} = 6$$

$$\frac{40 \times 40}{200} = 8$$

Critical region:-

$$\chi^2 \geq \chi^2_{0.05}(k-1)(r-1)$$

$$\chi^2 \geq \chi^2_{0.05}(4-1)(3-1)$$

$$\chi^2 \geq \chi^2_{0.05}(3)(2)$$

$$\chi^2 \geq \chi^2_{0.05}(6)$$

$$\chi^2 \geq 12.59$$

$$13.2463 \geq 12.59$$

Conclusion:-

Since the calculated value of Chi-Square (χ^2) is greater than the critical region. So, the null hypothesis (H_0) is reject and alternative hypothesis (H_1) is accepted.

② Given the following information

	Sample			
	1	2	3	4
Sample Size	4	6	7	3
Sample Mean	58	57	43	42
Sample biased Variance	10	30	6	9

Construct the analysis of variance table and test the hypothesis that population means are equal. Use 5% level of Significance.

Solution:-

Hypothesis:-

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ all the three means are equal.
 $H_1: \text{All the three means are not equal.}$

Level of Significance:-

$$\alpha = 0.05$$

Test - Statistics:-

$$F = \frac{S_b^2}{S_w^2}$$

Calculation:-

Calculate overall

means.

$$\text{Sample 1: } 4 \times 58 = 232$$

$$\text{Sample 2: } 6 \times 57 = 342$$

$$\text{Sample 3: } 7 \times 43 = 301$$

Sample 4: $3 \times 42 = 126$

overall means =

Total sum

Total no. of observation

$$= \frac{232 + 342 + 301 + 126}{4 + 6 + 7 + 3}$$

$$= \frac{1001}{20}$$

overall mean = 50.05

SS Between = $4(58 - 50.05)^2 + 6(57 - 50.05)^2 + 7(43 - 50.05)^2 + 3(42 - 50.05)^2$

$$BSS = 4(7.95)^2 + 6(6.95)^2 + 7(-7.05)^2 + 3(-8.05)^2$$

$$= 4(63.2025) + 6(48.3025) + 7(49.7025) + 3(64.8025)$$

$$= 252.81 + 289.815 + 347.9175 + 194.4075$$

BSS = 1084.95

$$\text{Within SS} = 10(4-1) + 30(6-1) + 6(7-1) + 9(3-1)$$

$$= 3(10) + 5(30) + 6(6) + 9(2)$$

$$= 30 + 150 + 36 + 18$$

$$\boxed{\text{Within SS} = 234}$$

Then the ANOVA Table.

Source of variation	d.f	SS	MS	Computed F-ratio
Between	(k-1) (4-1)	1084.95	361.65	24.72
Sample	3			
within	(n-k) (20-4)			
sample	16	234	14.625	
Total	19	1318.95		

Critical region:-

$$F \geq F_{0.05}(3, 9)$$

$$F \geq F_{0.05}(3, 9)$$

$$F \geq F_{0.05}(3, 9)$$

$$24.72 \geq 3.86$$

Conclusion:-

Since the calculated value of F is greater than

the Critical region. So, the null hypothesis (H_0) is reject and the alternative hypothesis (H_1) is accepted.

- ③ An industry engineer is investigating the effects of four assembly methods A, B, C, and D on the assembly time for color television components. 4 operators are selected for the study. The engineer Latin Square design.

Order of Assembly	operators			
	1	2	3	4
1	C10	PUBP	A7	B9
2	B7	C16	D11	A11
3	A5	B10	C6	D14
4	D8	A10	B11	C11

Analyze the data in the following table and draw appropriate conclusion for assembly methods.

Solution:-

Hypothesis:-

H_0 : All the four assembly methods ($\mu_A = \mu_B = \mu_C = \mu_D$) are equal.

H_1 : All the four assembly methods are not equal.

Level of significance:-

$$\alpha = 0.05$$

Test-Statistics:-

$$F = \frac{\text{MS For Treatments}}{\text{MS For Error}}$$

Calculation:-

order of assembly	Treatments				R_i, T_i	R_i^2, T_i^2	X_{ij}^2
	1	2	3	4			
1	10(100)	14(196)	7(49)	9(81)	40	1600	426
2	7(49)	16(256)	11(121)	11(121)	45	2025	547
3	5(25)	10(100)	6(36)	14(196)	35	1225	357
4	8(64)	10(100)	11(121)	11(121)	40	1600	406
C_j, T_j	30	50	35	45	160	6450	1736
C_j^2, T_j^2	900	2500	1225	2025	6650		↓
X_{ij}	238	652	327	519	1736	→ checked	

$$\begin{aligned} \text{Total SS} &= \sum X_{ij} - \frac{T^2}{rc} \\ &= 1736 - \frac{(160)^2}{16} \end{aligned}$$

$$= 1736 - \frac{25600}{16}$$

$$= 1736 - 1600$$

$$\boxed{TSS = 136}$$

$$Rowss = \frac{\sum T_i^2}{K} - C.F$$

$$= \frac{6450}{4} - 1600$$

$$= 1612.5 - 1600$$

$$\boxed{Rowss = 12.5}$$

$$Columns SS = \frac{\sum T_j^2}{K} - C.F$$

$$= \frac{6650}{4} - 1600$$

$$= 1662.5 - 1600$$

$$= 1662.5 - 1600$$

$$\boxed{column SS = 62.5}$$

$$Treatment SS = ?$$

The sum of all the value of A

$$= 5 + 10 + 7 + 11$$

$$= 33$$

The sum of all values of B

$$= 7 + 10 + 11 + 9$$

$$= 37$$

The sum of all values of C

$$= 10 + 16 + 6 + 11$$

$$= 43$$

The sum of all values of D

$$= 8 + 14 + 11 + 14$$

$$= 47$$

$$\text{Treatment SS} = \frac{(33)^2 + (37)^2 + (43)^2 + (47)^2}{4} - \text{C.F.}$$

$$= \frac{1089 + 1369 + 1849 + 2209}{4} - \text{C.F.}$$

$$= \frac{6516}{4} - 1600$$

$$= 1629 - 1600$$

$$\boxed{\text{Treatment SS} = 29}$$

$$\text{Error SS} = \text{TSS} - (\text{RSS} + \text{CSS} + \text{TSS})$$

$$= 136 - (12.5 + 62.5 + 29)$$

$$= 136 - 104$$

$$\boxed{\text{Error SS} = 32}$$

Source of variation	d.f	SS	MS	Computed -F
Row	$(k-1)$ 3	12.5	4.166	---
Columns	$(k-1)$ 3	62.5	20.833	---
Treatments	$(k-1)$ 3	29	9.666	$F = 1.8124$
Error	$(k-1)(k-2)$ 6	32	5.333	---
Total	15	136	---	---

Critical region:-

$$F_{0.05}(3,6)$$

$$F \geq 4.76$$

$$1.8124 \geq 4.76$$

Conclusion:-

Since the value of calculated is less than the critical region, it does not fall in critical region. So the null hypothesis (H_0) is accepted and the alternative hypothesis (H_1) is rejected.

LSD:-

$$= \frac{t_{0.05}(6)}{2} \sqrt{\frac{2(MSE)}{K}}$$

$$= \frac{t_{0.025}(6)}{2} \sqrt{\frac{2(5.333)}{4}}$$

$$= t_{0.025(6)} \sqrt{\frac{10.666}{4}}$$

$$= (2.447) \sqrt{2.6665}$$

$$= (2.447) (1.6329)$$

$$\boxed{LSD = 3.9957}$$

Arranging the mean yields for the four assembly ordered in ascending order.

$$= \frac{33}{4}, \frac{37}{4}, \frac{43}{4}, \frac{47}{4}$$

Means

$$= 8.25, 9.25, 10.75, 11.75$$

- ④ Estimate a regression line from the following data of heights x and weight w of 12 person

Height (x)	weight (w)
60	110, 135, 120
62	120, 140, 130, 135
64	150, 145
70	170, 185, 160

Test the hypothesis that population regression coefficient $\beta = 0$.

Solution

Hypothesis:-

$$\beta = 0$$

$$\beta \neq 0$$

Level of Significance:-

$$\alpha = 0.05$$

Test - Statistics:-

$$t = \frac{b}{S_b}$$

Calculation:-

$$t = \frac{b}{S_b}$$

$$S_b^2 = \frac{S_{Y.X}^2}{\sum (X - \bar{X})^2}$$

$$S_{Y.X}^2 = \frac{\sum (Y - \bar{Y})^2}{n - 2}$$

$$S_{Y.X} = \sqrt{\frac{\sum (Y - \bar{Y})^2}{n - 2}}$$

$$\sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$\sum (Y - \bar{Y}) = \sum Y^2 - a \sum Y - b \sum XY$$

Estimated regression line

$$= \hat{Y} = a + bx$$

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

calculate the values

X_i	Y_i	X_i^2	Y_i^2	XY
60	110	3600	12100	6600
60	135	3600	18225	8100
60	120	3600	14400	7200
62	120	3844	14400	7440
62	140	3844	19600	8680
62	130	3844	16900	8060
62	135	3844	18225	8370
64	150	4096	22500	9600
64	145	4096	21025	9280
70	170	4900	28900	11900
70	185	4900	34225	12950
70	160	4900	25600	11200
Total	766	49068	246100	109380

Estimated regression line

$$\hat{Y} = a + bx$$

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

put the value

$$= \frac{(12)(109380) - (766)(1700)}{(12)(49069) - (766)^2}$$

$$= \frac{1312560 - 1302200}{592816 - 586756}$$

$$= \frac{10360}{6060}$$

$$b = 5.03$$

$$\bar{y} = a + b\bar{x}$$

$$a = \bar{y} - b\bar{x}$$

$$a = \frac{1700}{12} - (5.03) \left(\frac{766}{12} \right)$$

$$= 141.6667 - (5.03)(63.833)$$

$$= 141.667 - 321.08$$

$$a = -179.41$$

$$\hat{y} = a + b(x)$$

$$\hat{y} = -179.41 + 5.03x$$

$$t = \frac{b}{s_b}$$

$$S_b^2 = \frac{S_{y \cdot x}}{\sum (x - \bar{x})^2}$$

$$S_{y \cdot x}^2 = \frac{\sum (y - \bar{y})^2}{n - 2}$$

$$S_{y \cdot x} = \sqrt{\frac{\sum (y - \bar{y})^2}{n - 2}} \quad \because \sum (y - \bar{y}) = \sum y^2 - a \sum y - b \sum xy$$

$$= \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

put the values

$$= \sqrt{\frac{246100 - (-179.41)(1700) - (5.03)(109380)}{12 - 2}}$$

$$= \sqrt{\frac{246100 + 304997 - 550181.4}{10}}$$

$$= \sqrt{\frac{915.6}{10}}$$

$$= \sqrt{91.56}$$

$$S_{y \cdot x} = 9.5687$$

$$S_b^2 = \frac{S_{y \cdot x}^2}{\sum (x - \bar{x})^2} = \frac{S_{y \cdot x}}{\sqrt{\sum (x - \bar{x})^2}}$$

$$S_b = \frac{9.5687}{\sqrt{\frac{\sum(X)^2 - (\sum X)^2}{n}}}$$

$$= \frac{9.5687}{\sqrt{\frac{49068 - (766)^2}{12}}}$$

$$= \frac{9.5687}{\sqrt{\frac{49068 - 586756}{12}}}$$

$$= \frac{9.568}{\sqrt{\frac{49068 - 48296.33}{12}}}$$

$$= \frac{9.568}{\sqrt{171.667}}$$

$$= \frac{9.568}{13.1022}$$

$$S_b = 0.7303$$

$$t = \frac{b}{S_b} = \frac{5.03}{0.7303} = 6.89$$

$$[t = 6.89]$$

Critical region:-

$$|t| \geq \frac{t_{0.05}(n-2)}{2}$$

$$|t| \geq t_{0.025}(12-2)$$

$$|t| \geq t_{0.025}(10)$$

$$|t| \geq 2.228$$

$$6.89 \geq 2.228$$

Conclusion:-

Since the calculated value of $t = 6.89$ is greater than the critical region and falls in critical region. So, the null hypothesis (H_0) is rejected and alternative hypothesis (H_1) is accepted.

- ⑤ Suppose that yield observation were taken at 7 different agricultural experimental stations across the country along with the rainfall and amount of fertilizer used. The following

is obtained:

Fertilizer x_1	1	2	3	4	5	6	7
Rainfall x_2	1	2	1	3	2	2	3
Yield y	4	5	5	7	6	7	8

Fit a multiple linear regression y and x_1 and x_2 .

Solution:-

Multiple linear regression equation

$$\hat{y} = a + b_1x_1 + b_2x_2$$

$$b_1 = \frac{(\sum x_1 y)(\sum x_2^2) - (\sum x_2 y)(\sum x_1 x_2)}{\sum (x_1^2)(\sum x_2^2) - (\sum x_1 x_2)}$$

$$b_2 = \frac{(\sum x_2 y)(\sum x_1^2) - (\sum x_1 y)(\sum x_1 x_2)}{\sum (x_1^2)(\sum x_2^2) - (\sum x_1 x_2)}$$

Calculate the value of b_1 and b_2 according to given table.

x_1	x_2	y	x_1^2	x_2^2	$x_1 y$	$x_2 y$	$x_1 x_2$
1	1	4	1	1	4	4	1
2	2	5	4	4	10	10	4
3	1	5	9	1	15	5	3
4	3	7	16	9	28	21	12
5	2	6	25	4	30	12	10
6	2	7	36	4	42	14	12
7	3	8	49	9	56	24	21
Total	28	42	140	32	185	90	63

$$b_1 = \frac{(\sum x_1 y)(\sum x_2^2) - (\sum x_2 y)(\sum x_1 x_2)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

$$= \frac{(185)(32) - (90)(63)}{(140)(32) - (63)^2}$$

$$= \frac{5920 - 5670}{4480 - 3969}$$

$$= \frac{250}{511}$$

$$b_1 = 0.49$$

$$b_2 = \frac{(\sum x_2 y)(\sum x_1^2) - (\sum x_1 y)(\sum x_1 x_2)}{(\sum x_1^2)(\sum x_2^2) - [\sum (x_1 x_2)]^2}$$

$$= \frac{(90)(140) - (185)(63)}{(140)(32) - (63)^2}$$

$$= \frac{12600 - 11655}{4480 - 3969}$$

$$= \frac{945}{511}$$

$$[b_2 = 1.85]$$

$$\hat{y} = a + b_1 x_1 + b_2 x_2$$

$$a = \bar{y} - b_1 \bar{x}_1 - b_2 \bar{x}_2$$

$$\bar{y} = \frac{\sum y}{n} = \frac{42}{7} = 6$$

$$\bar{x}_1 = \frac{\sum x_1}{n} = \frac{28}{7} = 4$$

$$\bar{x}_2 = \frac{\sum x_2}{n} = \frac{14}{7} = 2$$

$$a = 6 - (0.49)4 - 2(1.85)$$

$$a = 6 - 1.96 - 3.7$$

$$\boxed{a = 0.34}$$

Multiple linear regression
equation

$$\hat{y} = a + b_1x_1 + b_2x_2$$

$$\hat{y} = 0.34 + 0.49x_1 + 1.85x_2$$

